

# Tony Lin

[tonylin49ine@gmail.com](mailto:tonylin49ine@gmail.com) | <https://tonys-website.vercel.app> | [www.linkedin.com/in/tony-lin1](https://www.linkedin.com/in/tony-lin1)

## PROFESSIONAL EXPERIENCE

### **Deloitte, Software Engineer**

**Jul 2024 - Present**

- ❖ Engineered 70+ production REST API endpoints in Java end-to-end from system design through deployment across \$15B in supply chain contracts, enabling live delivery order tracking, a multithreaded email framework, and quote sheet management
- ❖ Built 44 reusable React components in TypeScript for an enterprise issue management platform supporting 8 report upload flows, chatbot page, and issue metrics visualization for 43K users currently in production
- ❖ Improved backend performance by 32% during peak loads through batch processing pipelines and stored procedure optimization in SQL Server
- ❖ Created 59 tests with JUnit, Jest, and Playwright achieving 62% total coverage, cutting post-release defects by 21%, and containerized services with Docker to improve production reliability and deployment consistency
- ❖ Owned and built an AI-powered graph generation end-to-end in Java using Claude Sonnet 4.5 to transform files with up to 2M tokens into visualizations in 2 minutes (vs. ~2 hours), with optimized metadata embedding and automated data type detection
- ❖ Designed a schema-driven Model Context Protocol tool-routing service with a modular pipeline to parse user intent, validate tool I/O, and dispatch tool calls that achieved ~80% answer accuracy across a large-scale chatbot system
- ❖ Built an async I/O based Python backend for automated report generation from multiple format inputs using Azure Blob Storage and AWS Textract, reducing generation latency by 60% and manual effort by 95%

### **Allied Microbiota, Bioinformatics Engineer**

**Aug 2023 - Jun 2024**

- ❖ Migrated 2 TB of bacterial genome data into Google Cloud Storage and analyzed the dataset with Google BigQuery to train a gene classification machine learning model, which achieved an 70% accuracy level at detecting bacterial strains capable of degrading carcinogens
- ❖ Created and automated ETL filtering pipelines in python for relevant proteins and biological pathways to increase large dataset processing throughput by 80%

### **Deloitte, Software Engineer Intern**

**Jun 2023 - Aug 2023**

- ❖ Implemented JSON file ingestion to automatically preprocess and validate client datasets for training custom models for an LLM accelerator tool to expedite training and deployment of client-specific LLMs by 3-6 months

### **CiCaidA | Fullstack Engineer**

**Jun 2022 - Jan 2023**

- ❖ Shipped real-time health telemetry features end-to-end across web and mobile in a fast-moving startup environment, iterating from minimal viable product to production across 61 nursing homes with no dedicated QA or ops support
- ❖ Finalist in the 2021 EMedic Global Clinical Technology Competition (Hong Kong) and awarded \$5K at WolfieTank for innovation in digital healthcare.

## PROJECT

### **Q&A Web Application | React, Express, Node, MongoDB**

**Feb 2023 - May 2023**

- ❖ Designed and implemented a React and Redux frontend with multiple interactive views, handling authentication flows, voting interactions, and real-time UI state updates, backed by a Node/MongoDB API layer with 23 REST endpoints and JWT authentication.

## EDUCATION

**Computer Science, Bachelor's of Science**

**GPA: 3.5**

## TECHNOLOGIES

**Languages:** Java, JavaScript, TypeScript, Python, C, C++, Go, SQL, Bash, PowerShell, Kotlin, Linux

**Databases and Tools:** SQL Server, PostgreSQL, Git, MongoDB, Node, React, Express, JUnit, NoSql, HTML

**Cloud:** S3, Lambda, RDS, DynamoDB, Step Functions, ECS, CloudFormation, SQS, Vertex AI, BigQuery, Docker, Kubernetes, Elasticsearch, Azure Blob Storage, Datadog, CI/CD, Github Actions

**Certifications:** U.S. Secret Clearance, AWS Cloud Solutions Architect, GCP Digital Cloud Leader